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Continuous-variable QKD over 50 km commercial fiber

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Abstract

The continuous-variable version of quantum key distribution (QKD) offers the advantages (over discrete-variable systems) of higher secret key rates in metropolitan areas, as well as the use of standard telecom components that can operate at room temperature. An important step in the real-world adoption of continuous-variable QKD is the deployment of field tests over commercial fibers. Here we report two different field tests of a continuous-variable QKD system through commercial fiber networks in Xi'an and Guangzhou over distances of 30.02 km (12.48 dB) and 49.85 km (11.62 dB), respectively. We achieve secure key rates two orders-of-magnitude higher than previous field test demonstrations by employing an efficient calibration model with one-time evaluation. This accomplishment is also realized by developing a fully automatic control system which stabilizes system noise, and by applying a rate-adaptive reconciliation method which maintains high reconciliation efficiency with high success probability in fluctuated environments. Our results pave the way for the deployment of continuous-variable QKD in metropolitan settings.

Introduction

Quantum key distribution (QKD) [1, 2] is one of the most practical applications in the field of quantum information. Its primary goal is to establish secure keys between two legitimate users, typically named Alice and Bob. Continuous-variable (CV) QKD [3, 4] has attracted much attention in the past few years, mainly as it uses standard telecom components that operate at room temperature and it has higher secret key rates (bits per channel use) over metropolitan areas [5–10]. We deployed the CV-QKD protocol based on coherent states [11–13] with Gaussian modulation that has been proven secure against arbitrary attacks [14–17]. Such attack is the most optimal in the asymptotical limit [18] and is also used in the finite-size regime [19–21]. Furthermore, many experimental demonstrations of CV-QKD protocol have also been achieved [6, 22–25].

Generally speaking, there are three basic criteria for a practical QKD system: automatic operation, stabilization under a real-world environment, and a moderate secure key rate [26]. Up to now, all previous long distance CV-QKD demonstrations have been undertaken in the laboratory without perturbation of the field environment [5, 27]. More recently, demonstrations based on CV quantum teleportation and CV Einstein–Podolsky–Rosen entangled states under labortory environment have been reported [28, 29]. The longest field tests of a CV-QKD system have been achieved over a 17.52 km deployed fiber (10.25 dB loss) [30] and a 17.7 km deployed fiber (5.6 dB loss) [31], where the secure key rates were 0.2 kbps and 0.3 kbps, respectively. Compared with field tests for discrete-variable QKD systems [32–36], these demonstrations have limited transmission distances and low key rates. The demonstration of field tests over longer metropolitan distances using CV-QKD has yet to be achieved.

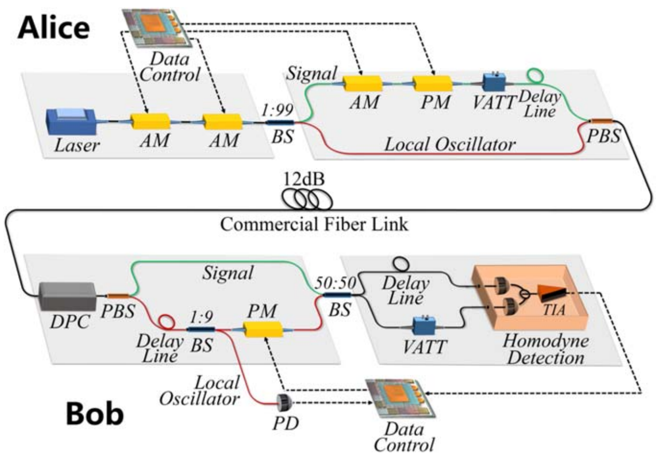

Figure 1. Optical layout for the field tests of the continuous-variable QKD system. Alice sends an ensemble of 40 ns Gaussian-modulated coherent states to Bob with a strong local oscillator multiplexed in time using a delay line and orthogonal polarization via a polarizing beamsplitter. The states are demultiplexed on Bob's side with another polarizing beamsplitter combined with an active dynamic polarization controller. After demultiplexing, the signal and local oscillator interfere on a shot-noise-limited balanced pulsed homodyne detector with a phase modulator on the local oscillator path to perform the random choice of the measured signal quadrature. Laser: continuous-wave laser; AM: amplitude modulator; PM: phase modulator; BS: beamsplitter; VATT: variable attenuator; PBS: polarizing beamsplitter; DPC: dynamic polarization controller; PD: photodetector.

There are several challenges to develop a practical CV-QKD system from a laboratory to the real world. Deployed commercial dark fibers are inevitably subject to much stronger perturbations due to changing environmental conditions and physical stress. This in turn causes disturbances of the transmitted quantum states. Deployed commercial dark fibers also suffer from higher losses due to splices, sharp bends and inter-fiber coupling. The software and hardware of CV-QKD modules must not only be designed to cope with all the conditions affecting the transmission fiber, but also must be robustly engineered to operate in premises designed for standard telecom equipment. Furthermore, as the system need to run continuously and without frequent attention, they should also be designed to automatically recover from any errors and shield the end users from service interruptions.

In this paper, we deploy CV-QKD protocol based on coherent states and achieve secure commercial fiber transmission distances of 30.02 km (12.48 dB loss) and 49.85 km (11.62 dB loss) in Xi'an and Guangzhou, two cities in China, respectively. The corresponding secret key rates are two orders-of-magnitude higher than that of previous demonstrations [30, 31, 37]. We achieve this by developing a fully automatic control system, that achieves stable excess noise, and also by applying a rate-adaptive reconciliation protocol to achieve a high reconciliation efficiency with high success probability. High secret key rate has been accomplished by using our new proposed one-time calibration model, which provides more efficient calibration procedure as well as simpler experimental implementations.

Results

The CV-QKD experimental setup is illustrated in figure 1 and consists of two legitimate users, Alice and Bob. To begin with, Alice generates 40 ns coherent light pulses by a 1550 nm telecom laser diode and two high-extinction amplitude modulators. Here the line width of the laser is 0.5 kHz and the extinction ratio of the amplitude modulators is 45 dB. These two amplitude modulators are pulsed with a duty cycle of 20% with a frequency of 5 MHz. These pulses are split into weak signals and strong local oscillators (LO) with a 1/99 beamsplitter, which ensures that the LO has enough power. The signal pulse is modulated with a centered Gaussian distribution using an amplitude and a phase modulator, where the modulation data is originally derived from our self-designed quantum random number generator [38, 39]. The variance is controlled using a variable attenuator and optimized with the loss of 12 dB. The signal pulse is delayed by 45 ns with respect to the LO pulse using a 9 m delay line. The devices used by Alice are polarization-maintaining. Both pulses are multiplexed with an orthogonal polarization using a polarizing beamsplitter. The time and polarization multiplexed pulses are sent to Bob in one fiber to reduce the phase noise caused in the transmission process.

After the fiber link, Bob first uses the dynamic polarization controller (DPC) to compensate the polarization drift, so that the polarization extinction ratio of polarization demultiplexing is kept at a high level. In our field tests, the extinction ratio of polarization after compensation is about 30 dB. A second delay line on Bob's side, which corresponds to the one used by Alice, allows for a time superposition of both the signal and LO pulses. A part of the LO, around 10%, is used in the system for clock synchronization, data synchronization, and LO monitor. A phase modulator on the LO path allows for a random selection of the measured signal quadrature

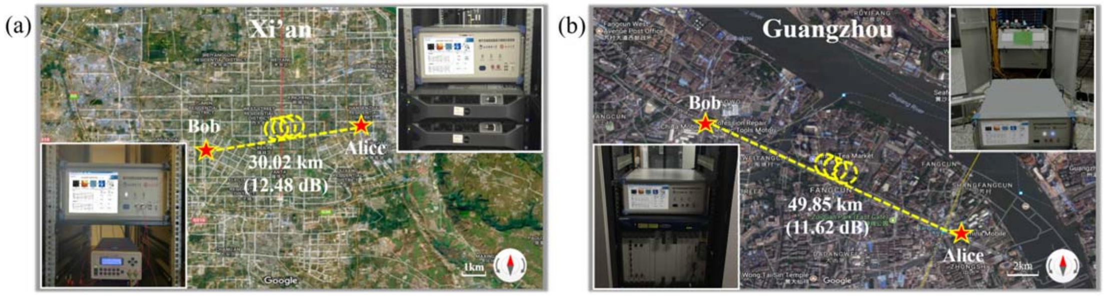

Figure 2. Bird's-eye view of the two field-environment CV-QKD systems. (a) Field test environment in Xi'an, where Alice is situated at a HuoJuLu Service Room (Xi'an Uotocom Network Technology Co., Ltd.) and Bob is situated at a HeShengJingGuang Service Room (Xi'an Uotocom Network Technology Co., Ltd.). The deployed fiber length is 30.02 km (12.48 dB). (b) Field test environment in Guangzhou, where Alice is at a QingHeDong Service Room (China Mobile Group Guangdong Co., Ltd.) and Bob is at a FangCun Service Room (China Mobile Group Guangdong Co., Ltd.). The deployed fiber length is 49.85 km (11.62 dB). ©Google Maps—2017.

and compensation of the phase drift between signal path and LO path. Then the signal and LO interfere on a shot-noise-limited balanced pulsed homodyne detector [40]. A time delay fiber and an attenuator are used to compensate the bias of the beamsplitter and the photodetectors.

For the fiber link, we undertake the field tests in the commercial fiber networks in two different cities in China. The first one is in Xi'an and operated by Xi'an Uotocom Network Technology Co., Ltd. The second one is in Guangzhou and operated by China Mobile Group Guangdong Co., Ltd. The channel loss of these two fiber links is similar, i.e. approximately 12 dB. However, the transmission distances are very different due to the different types of physical networks. As illustrated in figure 2(a), the first network is operated in the inner city of Xi'an, which has higher channel loss per kilometer. The total deployed fiber length is 30.02 km and the transmission loss is 12.48 dB for the fiber link (0.416 dB km^{-1}). The second network is between two different districts in Guangzhou, as shown in figure 2(b), the total deployed fiber length is 49.85 km and the transmission loss is 11.62 dB for the fiber link (0.233 dB km^{-1}). For the first field test, Alice is placed at the site of the HuoJuLu Service Room in Xi'an ($N 34^\circ 15' 12''$, $E 109^\circ 0' 44''$) and Bob at the site of the HeShengJingGuang Service Room ($N 34^\circ 14' 8''$, $E 108^\circ 53' 49''$). For the second field test, Alice is placed at the site of the QingHeDong Service Room in Guangzhou ($N 22^\circ 59' 20''$, $E 113^\circ 24' 9''$) and Bob at the site of the FangCun Service Room ($N 23^\circ 5' 49''$, $E 113^\circ 14' 8''$).

To overcome the channel perturbations due to changing environmental conditions, we developed several automatic feedback systems to calibrate time, polarization, and phase of the quantum states transmitted. For the time calibration shown in figure 1, there are two modules. One is a data synchronization module, and the other one is a clock synchronization module to achieve a synchronous clock in two remote places. Here we utilize the LO to perform the data synchronization as well as the clock synchronization. The traditional way involves using the modulated signal for the data synchronization module, which requires much more data to eliminate the effects of noise (the signal-to-noise ratio (SNR) of the signal is always extremely low (<1) when transmitted long distances) and the success possibility cannot reach 100%. For the polarization calibration, we adopt a DPC comprised of an electric polarization controller (EPC), a polarization beam splitter (PBS), and a photodetector.

With the use of this polarization calibration system operated in real time, the polarization mode is maintained and the power of the LO arm over the signal arm keeps over 30 dB. For the phase calibration, we utilize the phase modulator in the LO path (the same one as the basis sifting) to compensate the phase drift. Alice inserts the reference data in the signal sequence according to a certain period. The period of insertion is determined by the frequency of the phase drift. In the field tests, 100 reference data are inserted into every 1000 quantum signals to perform the phase estimation and compensation since the major contribution of the phase drift comes from the separate arms in Alice and Bob's MZI. The phase drift between the LO and the signal is obtained by the measurement results of the reference signal. Then the feedback voltage of the phase modulator is calculated from the half-wave voltage and is loaded onto the phase modulator in real time.

The shot-noise unit (SNU) can be vital to a CV-QKD system as the calibrated SNU will be treated as a normalization parameter to quantize the quadrature measurement results. According to the security analysis of CV-QKD, to evaluate the secret key rate we need to first resolve the corresponding covariance matrix, where its elements need to be normalised by the calibrated SNU. Therefore, the calibrated SNU will directly affect the security analysis on secret key rate. In these field tests the quantum signal path in the homodyne detection is randomly cut off to perform the SNU measurement.

After measurement, Alice and Bob will perform postprocessing to distill the final key. The postprocessing of a CV-QKD system contains four parts: basis sifting, parameter estimation, information reconciliation and privacy amplification. To support the high repetition frequency of a CV-QKD system, all parts of the

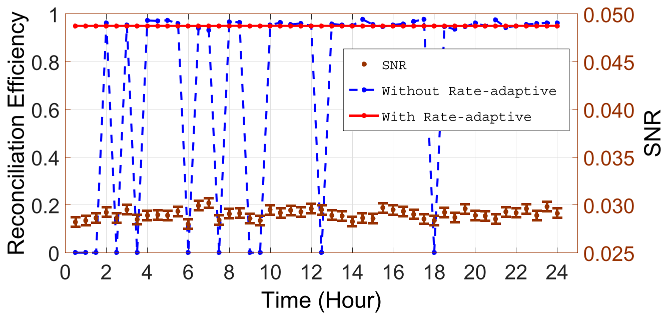

Figure 3. 24 h continuous test results for the signal-to-noise (SNR) ratio and reconciliation efficiency; in brown, the signal-to-noise-ratio of our system; in blue, the reconciliation efficiency without the rate-adaptive method [49]; in red, the reconciliation efficiency with a rate-adaptive method. The reconciliation efficiencies vary greatly without rate-adaptive method which is due to it being very sensitive to the variance of the SNRs of the quantum channel. The reconciliation efficiencies keep a high level with the rate-adaptive method.

postprocessing need to be executed at high speed. The computational complexity of basis sifting and parameter estimation is low. We can obtain high speed execution on a CPU, where the average speed of basis sifting and parameter estimation can be achieved up to 17.36 Mbits s^{-1} and 16.49 Mbits s^{-1}, respectively.

In the information reconciliation part, we combine multidimensional reconciliation and multi-edge type LDPC codes to achieve high efficiency at low SNRs [41, 42]. However, the speed of the error correction is limited on a CPU because of the long block code length and many iterations of the belief propagation decoding algorithm. We implement multiple code words decoding simultaneously based on our GPU and obtain speeds up to 30.39 Mbits s^{-1} on an NVIDIA TITAN Xp GPU [43]. Privacy amplification is implemented by a hash function (Toeplitz matrices in our scheme) [44, 45]. The average speed of privacy amplification can be achieved to 1.35 Gbps at any input length on the GPU [46].

Taking finite-size effects into account, the secret key rate, bounded by collective attacks, is given by [47]

$$K = f(1-\alpha)(1-\mathrm{FER})[\beta I(A{:}\ B) - \chi(B{:}\ E) - \Delta(n)], \tag{1}$$

where $\beta \in [0,\ 1]$ is the reconciliation efficiency, $I(A{:}\ B)$ is the classical mutual information between Alice and Bob, $\chi(B{:}\ E)$ is the Holevo quantity [48], $\Delta(n)$ is related to the security of the privacy amplification, FER is the frame error rate related to the reconciliation efficiency of a fixed error correction matrix, f is the repetition rate of a QKD system (5 MHz for our system) and α is the system overhead. The overhead here means the percentage of the signals that cannot be used to distill the final secret keys. In our system, some of the signal will be used to do the phase compensation.

To achieve a high key rate, we need to achieve a high reconciliation efficiency and have a low FER, which is a trade off in the experimental realization. For a fixed error correction matrix, achieving a high reconciliation efficiency will lead to a high FER. Thus, in the experiment, we need to find an optimal efficiency and an FER to maximize the secret key rate according to their relationship. Furthermore, we also need to decrease the system overhead to let more pulses distill the final keys.

For the high reconciliation efficiency and low FER part, we utilize the rate-adaptive reconciliation protocol [49]. As shown in figure 3, the reconciliation efficiency of decoding with the original MET-LDPC code [5, 49, 50] varies drastically (blue dash dotted line), because the practical SNRs of the quantum channel float in a range. However, rate-adaptive reconciliation [49] can keep high efficiencies even if the SNRs are unstable. For the parity check matrix with a code rate of 0.02, the maximum reconciliation efficiency of 97.99% can be achieved when the SNR is 0.0287. Although such high reconciliation efficiency can be achieved, the secret key rate is not necessarily high because the FER is also higher. According to equation (1), the secret key rate is influenced by both reconciliation efficiency and FER. Technically, the trend of the efficiency and the FER are opposite. Thus, we can find the optimal trade-off between efficiency and FER to maximize the secret key rate. Practically, for the parity check matrix with an original code rate of 0.02, we can obtain the maximum secret key rate when the practical SNR is 0.0296. Here the efficiency is 95% and the FER is 0.1. Thus, we use rate-adaptive reconciliation to maximize the secret key rate of our system by balancing the efficiency and the FER. Otherwise, the secret key rate will be reduced. Actually, it is difficult to find a fixed relationship between FER and reconciliation efficiency. Because it is related to many factors, including the degree distribution and construction methods of error correction codes, the distribution and randomness of the raw keys. The results in the paper are obtained by a fixed error correction code, and this code is chosen in particular.

For the system overhead part, we can swap the order of parameter estimation and information reconciliation to have an almost doubling of the final key rates [51, 52]. According to equation 1, the ratio of the data which is used to

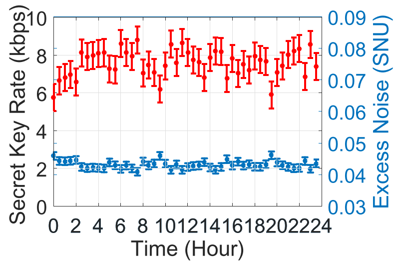

Figure 4. 24 h continuous test results for the secret key rate and the excess noise; lower blue mark is the excess noise; upper red mark represents the the secret key rate.

extract the secret keys has an important impact on the secret key rate. In the previous CV-QKD system, there are only n variables used to extract the keys. The other $m = N - n$ variables are used to estimate the quantum channel parameters. Taking into account the influence of the finite-size regime, the ratio of m to N is great for long distance CV-QKD systems. Generally, half of the data is used for parameter estimation, which will reduce the secret key rate by 50%. In our system, we swap the order of parameter estimation and information reconciliation. We use all the data to extract the keys, if Alice and Bob successfully correct the errors between them, Alice can recover Bob's raw keys. If the decoding fails, Bob discloses his raw keys. Therefore, no matter the decoding step successes or not, Alice can use the whole raw keys for parameter estimation. But still the failure is unlikely to happen. At the same time, since we use all of the raw keys to distill the secret keys, the secret key rate can be improved.

Using our CV-QKD system integrated with feedback systems, we have accumulated raw data for 24 h in Xi'an and 3 h in Guangzhou. During these periods, the automatic feedback systems worked effectively (details in appendix A). Compared with laboratory experiments, the field tests faced harsher environmental turbulence. For instance, the field environment will change the arrival time of the signals. The time calibration system works to monitor the time shift and then compensate it effectively. The achieved timing calibration precision is below 200 ps, which is much smaller than the 40 ns pulse width of the signal laser. Furthermore, with the help of the aforementioned polarization calibration system, we have compensated for the polarization change and achieved a fluctuation less than 5% in the Signal path. Finally, the achieved phase calibration precision is below 1°, according to the reference [27], the residue phase drift after the phase compensation causes approximately 0.0004 of the excess noise, which is relatively less compare to the total excess noise.

To one step further improving the experimental system, we propose and apply a new calibration model with one-time evaluation. Comparing to the rather complicated two-time calibration process, in the one-time evaluation model we only need to measure once the homodyne detector when the LO path is on, and take the result as the SNU. The new calibration model also attributes to simplify the system implementation as only one optical switch is demanded during the calibration procedure. Moreover, the statistical fluctuation of the SNU introduced by the calibration procedures is reduced. The channel loss versus secret key rate curves have been depicted in figure 5, the red solid curve represents the secret key rate in the asymptotic limits when applies our new proposed calibration model, the black dash-dot curve represents the secret key rate in the asymptotic limits but under the original calibration model. It can be observed that the performance of the one-time calibration model and the original two-times calibration model are almost the same except for a slightly different when the channel loss is over 20 dB.

The 24 h continuous test results in the Xi'an network for secret key rates and excess noises are shown in figure 4, where the average excess noise of our system is 4% SNU. It should be noticed that the composable security which can help building up a more complete analysis under a practical environment can be carried out in further work. The secret key rate is 7.57 kbps in the asymptotic limit, while the key rate is 5.91 kbps in finite-size regime (details can be found in appendix. E Table E1.). The average reconciliation efficiency is 0.9501 with an average SNR of 0.0295 in our continuous test in Guangzhou. The continuous test results in the Guangzhou network for the secret key rate is 7.43 kbps in the asymptotic limit and 5.77 kbps in the finite-size regime (details can be found in appendix. E Table E2.). Due to the feedback procedure, this CV-QKD system can run continuously for long periods of time automatically.

Discussion

With these field tests of a CV QKD system, we have extended the distribution distance to 50 km over commercial fiber [30, 31, 37]. In addition, with an optimized scheme in the optical layer and postprocessing, the secure key rates are higher than previous results by two orders-of-magnitude, which is now comparable to the key rates of discrete-variable

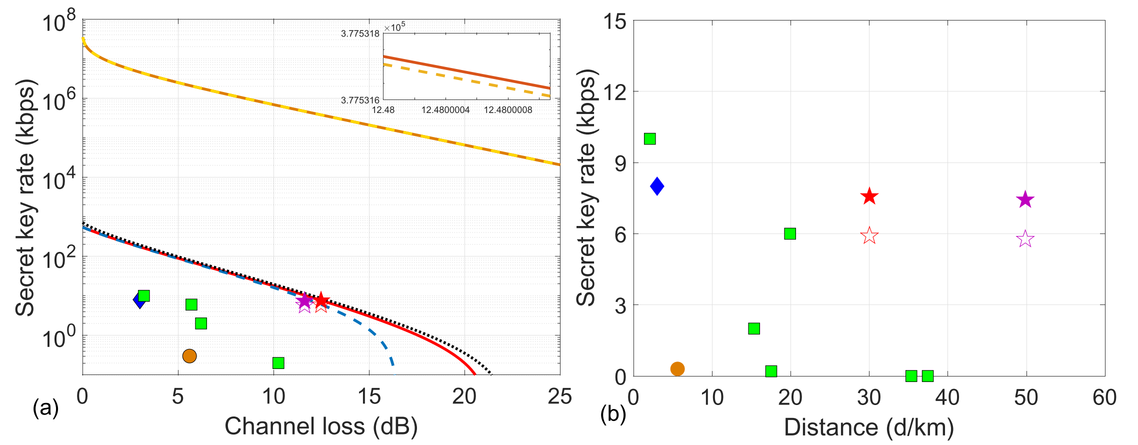

Figure 5. Secure key rates of experiments in the commercial fiber as well as the simulation results shown as a function of fiber loss and length. The red and purple five-pointed stars correspond to the experimental results in Xi'an and Guangzhou with the fiber transmitting losses of 12.48 dB (30.02 km) and 11.62 dB (49.85 km), respectively. Solid five-pointed stars are the experimental results under asymptotic limit while the hollow five-pointed stars represent the experimental results under finite-size regime. The black dot line shows the results with the original calibration model under asymptotic limit regime. The red solid curve represents the result calculated with the experimental parameters in the asymptotic limit under one-time-evaluation calibration model. The dashed blue curve represents the result in the finite-size regime with the one-time-evaluation calibration model. The PLOB bound which provides the fundamental limit of repeaterless quantum communications is also drawn in the figure. This is the brown–yellow line which corresponds to the PLOB bound for a lossy channel [53] and the one for a thermal-loss channel [53, 54] considering the average excess noise in our field test. The square points, rhombus point, and the dot point represent the field trials in [30, 31, 37].

QKD systems at metropolitan distances [32–36]. The PLOB bounds which show the maximum achievable secret key rate in QKD have also been drawn in figure 5 for both the lossy channel and thermal-loss channel. Although we have achieved highest secret key rates among the reported field tests, the secret key rates are still lower than the PLOB bounds for both the lossy channel and thermal-loss channel. The secret key rate we achieved here is based on the system with a repetition rate which is only 5 MHz. This repetition rate which is much smaller comparing to discrete-variable QKD systems (around 1 GHz). These results have moved CV QKD towards a more practical setting and the expectation that a secure metropolitan network could be built and is within reach of current technology.

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Additional information

Competing financial interests: The authors declare that they have no competing interests.

Appendix A. Experimental details

There are three automatic feedback systems to calibrate time, polarization, phase of the quantum states transmitted in our system in detail. These three feedback systems are the key modules to run a CV-QKD system in deployed commercial fibers.

For the time calibration shown in figure 1, there are two modules. One is data synchronization module, and another one is clock synchronization module to get synchronous clock in two remote places. For the data synchronization module, here we utilize a frame synchronization module based on the modulation of the LO. Previously, the traditional way is using the modulated signal, which requires much more data to eliminate the effects of noise (The SNR of Signal is always extremely low (<1) when transmitted long distance) and the success possibility can not reach 100%. We modulate data synchronization information on the LO, whose SNR is high enough to reduce the time of synchronization and promote the success possibility to 100%. The data synchronization is used to decide the starting point of the signal, so that the receiving system starts to work. It is the premise of the normal operation of the system. The data synchronization signal is a specific train sequence and modulated on the first high-extinction AM for the pulse generation. Because the clock is precisely synchronized in our system, the data synchronization is implemented with a low frequency (0.5 kHz) which introduce ignorable effect on shot-noise calibration and the key distribution. For the clock synchronization module, we also utilize LO to accomplish that. Bob utilizes a photoelectric detector (PD) to detect the part of LO pulses. The output signals of the PD is shaped by the comparator, the shaped pulses (5 MHz) are inputted into a clock chip to generate the high frequency clock (200 MHz) which is required by the other modules of the system. In the previous implementation [5], there is only the realization of clock synchronization by LO but no mention of the data synchronization.

For the polarization calibration, we adopt a polarization stabilization system comprised of an EPC, a PBS, and a photodetector. We insert the EPC and PBS after the fiber link, where the transmission port of PBS is connected to Lo path and the reflection port of the PBS is connected to Signal path. One percent of the power of LO pluses are utilized to monitor the polarization state and calculate the feedback voltage of EPC. With the use of this polarization calibration system operated in real time, the polarization mode is maintained and the power ratio of the LO and the Signal keeps 30 dB. The all-fiber construction of the calibration system provides very low insertion loss. Compared to the polarization compensation in [5], distilling the drift information needs more homodyne detection results than the scheme based on detecting the leaked LO in signal path. When the repetition frequency of the system is higher than polarization drift ratio, the amount of detection results is large enough to calculate drift. But in order to deal with various situations where polarization drift may be severe, the polarization controller based on the high-SNR LO pulses is more applicable.

For the phase calibration, although the LO and the Signal is transmitted in one fiber with a small delay, a slow phase shift which is aroused from the difference between the LO path and Signal path is unavoidable. To stabilize this, we utilize the phase modulator in the LO path (the same one as the base sifting) to compensate the phase drift. Alice inserts the reference data in the signal sequence according to a certain period. The period of insertion is determined by the frequency of the phase drift. The phase drift between the LO and the Signal is obtained by the measurement results of the reference signal. Then the feedback voltage of the phase modulator is calculated from the half-wave voltage and is loaded onto the phase modulator in real time.

Appendix B. Calibration model with one-time evaluation

In the previous experimental demonstrations, two-times calibration process is extensively used as it requires firstly measuring the homodyne detector output when both LO path and signal path are off, then the homodyne detector output when only LO path is connected. Finally using subtraction to calculate the SNU. In the proposed one-time-evaluation calibration model, we only need to measure once the homodyne detector when the LO path is on, thus the SNU of these two models can be deduced as:

$$\begin{aligned} \mathrm{SNU}^{\mathrm{TTE}} &= V_{\mathrm{tot}} - V_{\mathrm{ele}}, \\ \mathrm{SNU}^{\mathrm{OTE}} &= V_{\mathrm{tot}} = \mathrm{SNU}^{\mathrm{TTE}} + V_{\mathrm{ele}}, \end{aligned} \tag{2}$$

where V_{tot} and V_{ele} are corresponding to the variance of the output of the homodyne detector when the LO path is on and off. $\mathrm{SNU}^{\mathrm{TTE}}$ represents the two-times calibration SNU and $\mathrm{SNU}^{\mathrm{OTE}}$ represents the SNU with one-time calibration.

Under this modeling, certain advantages can be procured: firstly, we only need one optical switch in signal path in our system. Secondly, only one-time statistic evaluation is required, which makes it a more utility model applying in practical systems. Thirdly, since we only need one-time period to calculate the SNU, the statistic fluctuation is minimized compared to original calibration model with two-times evaluation.

We now give a more detailed analysis on its availability. We first consider the output of a practical homodyne detector with limited detection efficiency and electronic noise:

$$X_{\mathrm{out}} = AX_{\mathrm{LO}}(\sqrt{\eta_d}\hat{x}_B + \sqrt{1-\eta_d}\hat{x}_{v1}) + X_{\mathrm{ele}}, \tag{3}$$

where $\hat{x}_B$ represents the x-quadrature of the canonical components of mode after the transmission and $\hat{x}_{v_1}$ represents the vacuum state. X_{ele} is a Gaussian variable with variance v_{el}, A is the circuit amplification parameter. The output of the homodyne detector need to be further quantized by the SNU to estimate Eve's information. In two-times calibration model, it is $\mathrm{SNU}^{\mathrm{TTE}} = A^2X_{\mathrm{LO}}^2$. In one-time-evaluation model, it is $\mathrm{SNU}^{\mathrm{OTE}} = A^2X_{\mathrm{LO}}^2 + \langle X_{\mathrm{ele}}^2\rangle = A^2X_{\mathrm{LO}}^2 + v_{\mathrm{el}}$. Then the data used for postprocessing is:

$$x_{\mathrm{out}}^{\mathrm{OTE}} = \frac{AX_{\mathrm{LO}}}{\sqrt{A^2X_{\mathrm{LO}}^2 + v_{\mathrm{el}}}}(\sqrt{\eta_d}\hat{x}_B + \sqrt{1-\eta_d}\hat{x}_{v_1}) + \frac{\sqrt{v_{\mathrm{el}}}}{\sqrt{A^2X_{\mathrm{LO}}^2 + v_{\mathrm{el}}}}\hat{x}_{v_2}. \tag{4}$$

Here the Gaussian variable X_{ele} in equation (3) is replaced with a Gaussian operator $\sqrt{v_{\mathrm{el}}}\hat{x}_{v_2}$ since the electronic noise is not controlled by Eve, v_{el} is the variance of X_{ele} and $\hat{x}_{v_2}$ is the vacuum of variance 1. If let $\eta_e = \frac{A^2X_{\mathrm{LO}}^2}{A^2X_{\mathrm{LO}}^2 + v_{\mathrm{el}}}$, it can adequately characterize the electronic noise using a beamsplitter.

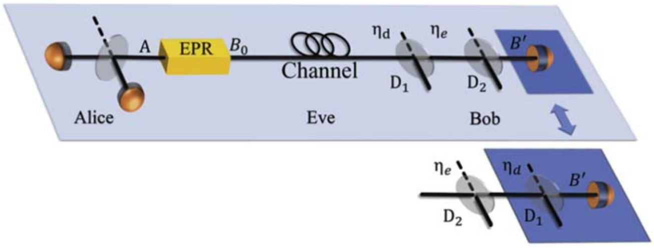

Figure B1. Entanglement-based model of one-time calibration model. Two beamsplitters are brought in to imitate the limited detection efficiency and the detector electronic noise. EPR states correspond to the two mode squeezed states that has two modes A and B, mode A is measured by Alice using heterodyne detection and mode B is sent to Bob. The mode arrived before the ideal homodyne detection is B'.

The completely version of the EB model is depicted in the upper side of figure B1. Alice prepares the EPR state that has two modes A and B where mode A is kept in Alice's side and will be measured by heterodyne detection. Mode B will be sent into the channel, in which Eve can conduct any attack according to her strategies. After the transmission, mode B will pass through two beamsplitters and become mode B' before the ideal homodyne detection. Based on the trusted model assumption, the limited detcted efficiency and electronic noise is not controlled by Eve thus if we commute the order of the beamsplitters it will not change the final detected mode B', the EB model after the beamsplitter switching is depicted on the bottom of figure B1. Under this modelling, we no longer need to measure the electronic noise.

Appendix C. Secret key rate with one-time-evaluation calibration model

The (asymptotical) secret key rate K with one-time-evaluation calibration model against collective attacks for reverse reconciliation is given by [55]

$$K = f(1 - \alpha)(1 - \mathrm{FER})[\beta I(A: B) - \chi(B: E)]. \tag{5}$$

The classical information of Alice and Bob $I(A: B)$ can be described by Shannon entropy which can be calculated by the variance of Alice and Bob and their covariance:

$$I(A: B) = \frac{1}{2}\log_2\left(\frac{V + \chi}{\chi + 1}\right). \tag{6}$$

Here we define χ as $\chi = \frac{1}{T\eta_d\eta_e} - 1 + \varepsilon_c$ for conciseness, T is the channel transmittance and ε_c stands for the channel excess noise.

Applying one-time-evaluation model, we no longer need to measure the electronic noise, so mode D_2 in the figure 5 is unknown to us. To simplify our calculation, however, the co-variance matrix of mode A, D_1, B' can be easily obtained:

$$\gamma_{AD_1B'} = \begin{pmatrix} \gamma_A & \phi_{AD_1} & \phi_{AB'} \\ \phi_{AD_1}^T & \gamma_{D_1} & \phi_{D_1B'} \\ \phi_{AB'}^T & \phi_{D_1B'}^T & \gamma_{B'} \end{pmatrix}. \tag{7}$$

Thus we write the Holevo quantity [48] $\chi(B: E)$ which restricts the upper bound of the information that the eavesdropper Eve can acquire as:

$$\chi(B: E) = \chi(B: E, D_2) = S(\rho_{AD_1B'}) - S(\rho_{AD_1}^{m_{B'}}). \tag{8}$$

The first term of the right side of the equation can be calculated by its corresponding co-variance matrix $\gamma_{AD_1B'}$ using its symplectic eigenvalues, the covariance matrix is derived as:

$$\gamma_{AB'D_1} = \begin{pmatrix} VI_2 & \sqrt{T\eta_e\eta_d(V^2-1)}\,\sigma_z & \sqrt{T\eta_e(1-\eta_d)(V^2-1)}\,\sigma_z \\ \sqrt{T\eta_e\eta_d(V^2-1)}\,\sigma_z & [T\eta_e\eta_d(V-1+\varepsilon_c)+1]I_2 & \sqrt{\eta_d(1-\eta_d)}\,T\eta_e(V-1+\varepsilon_c)I_2 \\ \sqrt{T\eta_e(1-\eta_d)(V^2-1)}\,\sigma_z & \sqrt{\eta_d(1-\eta_d)}\,T\eta_e(V-1+\varepsilon_c)I_2 & [T\eta_e(1-\eta_d)(V-1+\varepsilon_c)+1]I_2 \end{pmatrix}. \tag{9}$$

Experimentally, using Alice's data and Bob's data is sufficiently to obtain the parameter values appeared in the co-variance matrix above. The Von Neumann entropy $S(\rho_{AC}^{m_{B'}})$ is calculated from matrix $\gamma_{AD_1}^{m_{B'}}$, which is the matrix after Bob performs homodyne detection on mode B', it can be derived as:

$$\gamma_{AD_1}^{m_{B'}} = \gamma_{AD_1} - \phi_{AD_1B'}(X\gamma_{B'}X)^{MP}\phi_{AD_1B'}^{T}. \tag{10}$$

From co-variance matrix $\gamma_{AB'D_1}$ we can find three valid symplectic eigenvalues and from $\gamma_{AD_1}^{m_{B'}}$ we can find two valid symplectic eigenvalues. The $\chi(B\colon E)$ can now rewritten as:

$$\chi(B\colon E) = \sum_{i=1}^{3} G\left(\frac{\lambda_i - 1}{2}\right) - \sum_{i=4}^{5} G\left(\frac{\lambda_i - 1}{2}\right). \tag{11}$$

Where $G(x) = (x+1)\log_2(x+1) - x\log_2 x$. Then the final secret key rate is obtainable by using the equation (5). It should be noted that this calculation will lead to a secure lower bound of the actual secret key rate.

Appendix D. Rate-adaptive reconciliation protocol

The rate-adaptive reconciliation protocol is to maximize the secret key rate of our system by balancing the efficiency β and FER. With the combination of puncturing and shortening techniques, the code rate will be changed to

$$R = \frac{n - m - s}{n - p - s}, \tag{12}$$

where the original code rate is $R^o = (n - m)/n$, n is the length of the code and m is the length of redundancy bits. The code rate will be changed to $R' = (n - m)/(n - p)$ by adding punctured bits of length p. The code rate will be changed to $R'' = (n - m - s)/(n - s)$ by adding shortened bits of length s.

Therefore, we could have the reconciliation efficiency, which is defined as follows:

$$\beta = \frac{R}{C}, \tag{13}$$

where R is the rate of MET-LDPC code, C is the classical capacity of the quantum channel, which is $C = \frac{1}{2}\log_2(1 + \mathrm{SNR})$ for Gaussian variables.

The detailed steps of the rate-adaptive protocol are as follows [49]:

Step 1: According to the practical SNR of the experimental data (0.028 0.030), we calculate the optimal code rate. Then we select a good-performance original MET-LDPC code 0.02 whose code rate is close to the optimal code rate.

Step 2: We creates a new sequence $\hat{u}$ with length n in Bob's side:

$$u = \{u_1, u_2, \cdots\cdots, u_{n-p-s}\}, \tag{14}$$

$$p_B = \{p_{B1}, p_{B2}, \cdots\cdots, p_{Bp}\}, \tag{15}$$

$$s_B = \{s_{B1}, s_{B2}, \cdots\cdots, s_{Bs}\}, \tag{16}$$

$$\hat{u} = \{u_1, u_2, \cdots, s_{B1}, \cdots, p_{B1}, \cdots, s_{Bi}, \cdots, p_{Bj}, \cdots, u_k, \cdots\}, \tag{17}$$

where u is the string for Bob's multidimensional reconciliation with length $n - p - s$, the sequences p_B and s_B represent Bob's punctured bits and shortened bits which are randomly generated by Bob with length p and s, and $i \in \{1, 2, \cdots, s\}$, $j \in \{1, 2, \cdots, p\}$, $k \in \{1, 2, \cdots, n - p - s\}$. The new string $\hat{u}$ is created by randomly inserting p_B and s_B into the string u. Then Bob calculates the syndrome of $\hat{u}$, such that $c(\hat{u}) = H\hat{u}^T$. H matrix is the parity check matrix of the code, which is used to assist in the encoding and decoding process. Its rows refer to the check nodes and columns refer to the bit nodes. The nodes are obtained by the degree distribution. And H matrix is obtained by some construction method according to the degree distribution.

Step 3: After we receives the message sent by Bob in Alice's side, we constructs a new sequence $\hat{v}$ with length n:

$$v = \{v_1, v_2, \cdots\cdots, v_{n-p-s}\}, \tag{18}$$

$$p_A = \{p_{A1}, p_{A2}, \cdots\cdots, p_{Ap}\}, \tag{19}$$

$$\hat{v} = \{v_1, v_2, \cdots, s_{B1}, \cdots, p_{A1}, \cdots, s_{Bi}, \cdots, p_{Aj}, \cdots, v_k, \cdots\}, \tag{20}$$

where v is the sequence for Alice's multidimensional reconciliation with length $n - p - s$, the sequence p_A represents Alice's punctured bits which is randomly generated by Alice, and $i \in \{1, 2, \cdots, s\}$, $j \in \{1, 2, \cdots, p\}$, $k \in \{1, 2, \cdots, n - p - s\}$. The new string $\hat{v}$ and $\hat{u}$ have the same positions and length of punctured bits and shortened bits. And they also have the same value of shortened bits s_B. Then Alice uses the belief propagation decoding algorithm or some other algorithm to recover $\hat{u}$. Finally Alice and Bob will share a common string.

Appendix E. The detailed results of the field test in Xi'an and Guangzhou commercial fiber networks

Table E1. The detailed results of the field test in Xi'an commercial fiber network for 24 h. The detection efficiency of the homodyne detector is 0.612. SNR: practical signal-to-noise ratio. β^o: reconciliation efficiency of the original code. β: reconciliation efficiency of the rate-adaptive reconciliation. s: the length of shortened bits. p: the length of punctured bits. ϵ: excess noise. η_e: electronic noise $k_{\mathrm{Asymptotic}}$: final secret key rate in asymptotic limits. k_{finite}: finial secret key rate in finite-size regime.

Time	SNR	β^o	β	s	p	ϵ	ν_{el}	$k_{\mathrm{Asymptotic}}$ (kbps)	k_{finite} (kbps)
0:30	0.028 205	0	0.950 268	952	0	0.045 909	0.257 087 972	5.768 156 006	4.109 732 1
1:00	0.028 365	0	0.950 043	848	0	0.044 316	0.248 057 73	6.647 619 698	4.989 195 793
1:30	0.028 628	0	0.950 305	664	0	0.044 234	0.212 818 655	6.811 478 575	5.153 054 669
2:00	0.029 241	0.961 982	0.950 289	248	0	0.044 372	0.218 014 447	6.966 862 114	5.308 438 208
2:30	0.028 632	0	0.950 174	664	0	0.044 643	0.229 307 098	6.583 656 263	4.925 232 357
3:00	0.029 504	0.953 529	0.950 165	72	0	0.042 427	0.221 973 702	8.126 644 238	6.468 220 332
3:30	0.028 497	0	0.950 352	752	0	0.042 061	0.136 639 115	7.927 945 618	6.269 521 712
4:00	0.028 918	0.972 573	0.950 069	472	0	0.042 218	0.139 557 273	7.992 647 763	6.334 223 858
4:30	0.028 968	0.970 918	0.950 357	432	0	0.042 114	0.136 224 858	8.093 704 911	6.435 281 006
5:00	0.028 893	0.973 403	0.950 115	488	0	0.041 95	0.139 011 213	8.131 799 003	6.473 375 098
5:30	0.029 327	0.959 201	0.950 175	192	0	0.043 854	0.041 323 934	7.272 916 672	5.614 492 767
6:00	0.028 012	0	0.950 024	1088	0	0.042 957	0.040 551 421	7.229 510 101	5.571 086 196
6:30	0.029 93	0.940 153	0.949 994	0	10368	0.041 81	0.052 974 806	8.632 146 63	6.973 722 724
7:00	0.030 191	0.932 144	0.949 996	0	10360	0.042 883	0.051 965 41	8.137 967 712	6.479 543 807
7:30	0.028 433	0	0.950 132	800	0	0.041 958	0.112 702 51	7.938 746 112	6.280 322 206
8:00	0.029 086	0.967 035	0.950 35	352	0	0.040 907	0.111 844 32	8.802 540 33	7.144 116 424
8:30	0.029 145	0.965 105	0.950 346	312	0	0.044 158	0.142 472 731	7.051 915 692	5.393 491 787
9:00	0.028 574	0	0.950 145	704	0	0.042 901	0.148 243 261	7.491 192 309	5.832 768 403
9:30	0.028 368	0	0.950 333	840	0	0.043 532	0.110 357 487	7.088 975 27	5.430 551 365
10:00	0.029 49	0.953 975	0.950 235	80	0	0.045 93	0.109 311 065	6.204 816 116	4.546 392 21
10:30	0.029 18	0.963 964	0.950 357	288	0	0.043 481	0.159 636 778	7.434 540 248	5.776 116 342
11:00	0.029 437	0.955 668	0.950 048	120	0	0.041 518	0.154 011 192	8.590 946 155	6.932 522 249
11:30	0.029 28	0.960 719	0.950 171	224	0	0.043 213	0.156 899 036	7.604 695 583	5.946 271 677
12:00	0.029 627	0.949 627	0.95	0	392	0.041 517	0.160 396 589	8.668 259 572	7.009 835 666
12:30	0.028 19	0	0.950 375	960	0	0.042 405	0.145 677 068	8.150 250 582	6.491 826 677
13:00	0.029 35	0.958 46	0.950 193	176	0	0.042 722	0.154 258 455	7.735 690 531	6.077 266 626
13:30	0.029 502	0.953 593	0.950 228	72	0	0.042 887	0.146 898 664	7.621 970 71	5.963 546 804
14:00	0.029 6	0.950 481	0.950 108	8	0	0.043 946	0.147 434 369	6.813 599 789	5.155 175 883
14:30	0.028 777	0.977 271	0.950 056	568	0	0.042 241	0.204 325 891	7.871 299 729	6.212 875 823
15:00	0.029 437	0.955 668	0.950 048	120	0	0.041 536	0.199 646 968	8.219 492 648	6.561 068 742
15:30	0.029 704	0.947 201	0.949 998	0	2944	0.042 426	0.168 286 976	8.194 749 06	6.536 325 154
16:00	0.029 493	0.953 879	0.950 14	80	0	0.044 886	0.178 200 272	6.767 835 413	5.109 411 508
16:30	0.029 338	0.958 846	0.950 2	184	0	0.042 852	0.193 986 108	7.827 893 557	6.169 469 651
17:00	0.029 028	0.968 94	0.950 321	392	0	0.044 096	0.198 206 25	7.038 879 834	5.380 455 928
17:30	0.028 786	0.976 97	0.950 147	560	0	0.042 854	0.187 691 506	7.501 930 063	5.843 506 157
18:00	0.028 316	0	0.950 104	880	0	0.043 287	0.188 699 394	7.208 032 562	5.549 608 656
18:30	0.029 687	0.947 736	0.949 993	0	2376	0.042 483	0.112 187 631	7.968 544 83	6.310 120 924
19:00	0.030 07	0.935 839	0.949 998	0	14904	0.042 464	0.109 873 08	7.762 217 726	6.103 793 821
19:30	0.029 721	0.946 667	0.949 996	0	3504	0.043 288	0.321 548 662	7.672 138 455	6.013 714 549
20:00	0.029 262	0.961 301	0.950 371	232	0	0.046 126	0.338 690 481	5.903 249 913	4.244 826 007
20:30	0.029 622	0.949 785	0.949 998	0	224	0.043 842	0.144 982 226	7.088 051 03	5.429 627 124
21:00	0.028 868	0.974 234	0.950 162	504	0	0.042 965	0.143 486 744	7.437 802 729	5.779 378 823
21:30	0.028 632	0	0.950 17	664	0	0.044 643	0.207 438 908	8.033 042 141	6.374 618 236
22:00	0.029 504	0.953 529	0.950 16	72	0	0.042 427	0.204 039 996	8.223 936 63	6.565 512 724
22:30	0.028 497	0	0.950 35	752	0	0.042 061	0.321 594 474	8.349 520 597	6.691 096 691
23:00	0.028 918	0.972 573	0.950 07	472	0	0.042 218	0.338 761 392	6.861 580 907	5.203 157 002
23:30	0.029 83	0.943 258	0.949 999	0	7096	0.041 853	0.256 120 393	8.566 343 62	6.907 919 714
0:00	0.029 605	0.950 323	0.950 323	0	0	0.043 515	0.261 181 193	7.399 916 031	5.741 492 125

Table E2. The detailed results of the field test in Guangzhou commercial fiber network for continuous 48 blocks. The detection efficiency of the homodyne detector is 0.612. SNR: practical signal-to-noise ratio. s: the length of shortened bits. p: the length of punctured bits. β: reconciliation efficiency of the rate-adaptive reconciliation. β^o: reconciliation efficiency of the original code. η_e: electronic noise. ϵ: excess noise. $k_{\text{Asymptotic}}$: final secret key rate in asymptotic limits. k_{finite}: finial secret key rate in finite-size regime.

Time	SNR	β^o	β	s	p	ϵ	v_{el}	$k_{\text{Asymptotic}}$ (kbps)	k_{finite} (kbps)
1	0.030 241	0.930 625	0.949 997	0	20 392	0.042 136	0.392 581 912	8.579 503 464	6.921 079 558
2	0.030 869	0.911 972	0.949 994	0	40 024	0.041 306	0.364 250 03	9.320 395 083	7.661 971 177
3	0.030 093	0.935 135	0.949 992	0	15 640	0.042 493	0.399 424 436	8.316 653 184	6.658 229 278
4	0.028 783	0.977 07	0.950 245	560	0	0.044 921	0.463 095 576	6.496 388 751	4.837 964 845
5	0.030 063	0.936 054	0.949 992	0	14 672	0.042 497	0.400 822 547	8.302 073 854	6.643 649 948
6	0.029 988	0.938 361	0.949 996	0	12 248	0.042 856	0.404 315 349	8.070 705 358	6.412 281 452
7	0.030 522	0.922 184	0.95	0	29 280	0.042 205	0.379 741 876	8.658 056 183	6.999 632 277
8	0.030 833	0.913 02	0.949 994	0	38 920	0.042 003	0.365 815 545	8.903 768 839	7.245 344 933
9	0.030 322	0.928 176	0.949 995	0	22 968	0.042 875	0.388 825 962	8.195 404 522	6.536 980 616
10	0.029 675	0.948 113	0.949 998	0	1984	0.043 563	0.419 116 752	7.553 054 985	5.894 631 079
11	0.029 938	0.939 905	0.949 998	0	10 624	0.043 19	0.406 649 656	7.864 232 652	6.205 808 746
12	0.029 998	0.938 053	0.95	0	12 576	0.043 012	0.403 840 032	7.987 857 62	6.329 433 714
13	0.028 429	0	0.950 264	800	0	0.045 778	0.481 301 599	5.914 846 678	4.256 422 772
14	0.028 628	0	0.950 305	664	0	0.045 419	0.471 006 441	6.179 197 639	4.520 773 733
15	0.029 767	0.945 226	0.949 998	0	5024	0.043 204	0.414 740 19	7.788 255 159	6.129 831 253
16	0.030 184	0.932 357	0.949 996	0	18 568	0.042 452	0.395 201 719	8.377 533 14	6.719 109 234
17	0.030 235	0.930 807	0.949 997	0	20 200	0.042 501	0.392 843 225	8.371 020 725	6.712 596 819
18	0.030 251	0.930 322	0.949 998	0	20 712	0.042 284	0.392 114 741	8.500 176 807	6.841 752 902
19	0.028 431	0	0.950 198	800	0	0.045 295	0.481 217 627	6.165 721 806	4.507 297 901
20	0.030 509	0.922 571	0.949 999	0	28 872	0.042 212	0.380 330 252	8.648 552 53	6.990 128 624
21	0.030 307	0.928 628	0.95	0	22 496	0.042 723	0.389 520 645	8.275 546 956	6.617 123 051
22	0.030 149	0.933 423	0.949 999	0	17 448	0.042 298	0.396 829 981	8.450 038 968	6.791 615 062
23	0.030 778	0.914 628	0.949 998	0	37 232	0.041 647	0.368 274 421	9.085 252 889	7.426 828 984
24	0.030 286	0.929 263	0.949 995	0	21 824	0.042 745	0.390 484 445	8.254 117 209	6.595 693 303
25	0.028 312	0	0.950 236	880	0	0.045 654	0.487 436 318	5.936 894 638	4.278 470 733
26	0.029 887	0.941 485	0.949 997	0	8960	0.042 907	0.409 065 042	8.001 486 009	6.343 062 103
27	0.028 648	0	0.950 036	656	0	0.045 069	0.469 992 91	6.350 759 686	4.692 335 781
28	0.028 112	0	0.950 226	1016	0	0.046 306	0.498 004 738	5.524 620 815	3.866 196 909
29	0.030 052	0.936 392	0.949 996	0	14 320	0.042 57	0.401 332 871	8.256 950 665	6.598 526 759
30	0.029 944	0.939 719	0.949 995	0	10 816	0.043 148	0.406 369 206	7.889 801 849	6.231 377 943
31	0.028 482	0	0.950 07	768	0	0.045 612	0.478 548 565	6.005 455 74	4.347 031 835
32	0.028 341	0	0.950 057	864	0	0.046 096	0.485 893 688	5.699 515 506	4.041 091 601
33	0.028 639	0	0.950 331	656	0	0.045 393	0.470 441 801	6.199 128 217	4.540 704 311
34	0.028 151	0	0.950 105	992	0	0.046 136	0.495 933 886	5.617 077 248	3.958 653 342
35	0.028 581	0	0.950 301	696	0	0.045 512	0.473 424 664	6.112 615 37	4.454 191 464
36	0.028 332	0	0.950 354	864	0	0.045 976	0.486 371 377	5.783 854 353	4.125 430 447
37	0.028 073	0	0.950 348	1040	0	0.046 308	0.500 088 447	5.520 168 261	3.861 744 355
38	0.029 854	0.942 511	0.949 997	0	7880	0.043 236	0.410 610 721	7.805 029 83	6.146 605 924
39	0.030 517	0.922 333	0.949 996	0	29 120	0.042 385	0.379 960 638	8.553 215 1	6.894 791 194
40	0.028 003	0	0.950 325	1088	0	0.046 221	0.503 846 816	5.539 809 861	3.881 385 956
41	0.028 492	0	0.950 129	760	0	0.045 8	0.478 021 028	5.914 225 49	4.255 801 584
42	0.028 991	0.970 159	0.950 375	416	0	0.044 721	0.452 593 235	6.691 442 03	5.033 018 124
43	0.028 27	0	0.950 066	912	0	0.045 812	0.489 642 361	5.825 124 611	4.166 700 705
44	0.030 922	0.910 432	0.949 998	0	41 648	0.041 434	0.361 903 336	9.269 858 261	7.611 434 355
45	0.030 83	0.913 108	0.949 998	0	38 832	0.041 848	0.365 955 157	8.991 932 879	7.333 508 973
46	0.029 816	0.943 695	0.949 995	0	6632	0.043 106	0.412 416 326	7.861 992 735	6.203 568 83
47	0.028 779	0.977 204	0.950 375	560	0	0.045	0.463 295 864	6.463 372 889	4.804 948 983
48	0.030 125	0.934 156	0.949 994	0	16 672	0.042 621	0.397 930 61	8.258 044 941	6.599 621 035

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